

Study area

The Alps (Fig. 1A) are the centre of biodiversity in Europe hosting ca. 4,500 plant species (Chemini & Rizzoli, 2003; Theurillat, 1995) along its ~1,200 km length and ~400 km in width encompassing eight Countries. Their mountains are extraordinarily morphologically distinct with their lowest altitude recorded at sea level and their highest peak at 4,808 m.a.s.l., the Mont Blanc (at the border between Italy and France). The geology is particularly complex composed by many different topologies (i.e., sedimentary, calcareous and siliceous outcrops, ultramafic and ophiolites rocks; Hartmann & Moosdorf, 2012) and they represent a climatic “crossroad” that intersect oceanic, continental, mediterranean, polar and, on occasion, Saharan influence (Beniston, 2006). Such geological and climatic configuration is of fundamental importance for the diversity of plant species and the distribution of their habitats (Nagy et al., 2003). For the purpose of our study, we considered the orographic

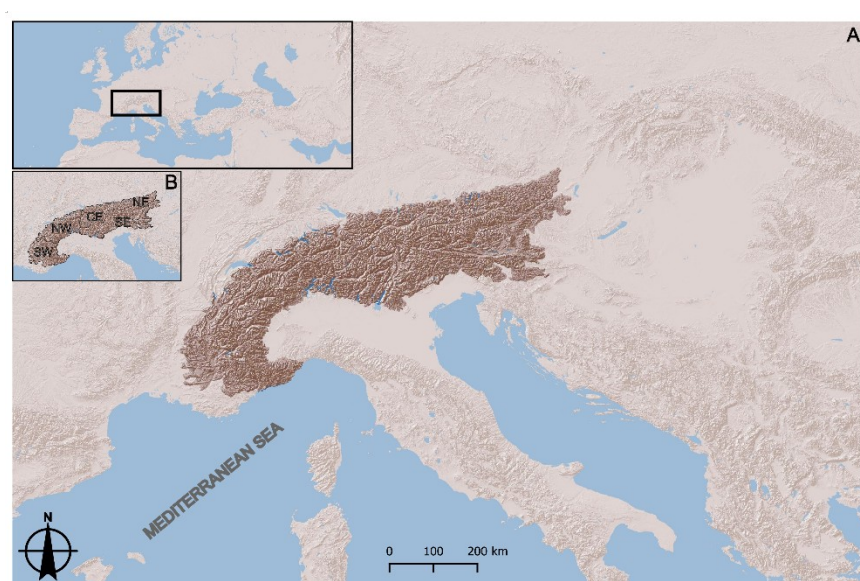


Figure 1

subdivision of the Alps into five sectors (south-west (SW), north-west (NW), north-

east (NE), central-east (CE) and south-east (SE); Marazzi, 2005) to account for regional climate variability of species-specific response to climate change (Fig. 1B).

Habitat selection

We selected ten dry, alpine and subalpine grassland habitats at level 3 of the classification hierarchy of the EUNIS Habitat Classification system (Chytrý et al., 2020) summarized in Table 1. According to their distribution occurrence in the Alps, these habitats were chosen as case studies representing different climatic and geological characteristics of the area altogether.

For each habitat we retrieved species with occurrences assembled in the habitat and rare or absent in others (i.e., diagnostic species *sensu* Westhoff and van der Maarel, 1973), for a total of 167 species, to calculate Species Distribution Models (SDMs) under future climates. Diagnostic species are more strongly associated with the habitat type (Whittaker, 1962) and generally perform better in SDMs than non-diagnostic species (e.g., dominant) as their distribution is strongly related to specific environmental conditions (Sumbh et al., 2025).

	<i>EUNIS Code</i>	<i>Habitat</i>	<i>Description (sensu Chytrý et al., 2020)</i>
	R12	Cryptogam- and annual-dominated vegetation on siliceous rock outcrops	Open pioneer grasslands with perennial succulents and spring annuals with subordinate small tussock grasses on siliceous outcrops

DRY GRASSLANDS	R13	Cryptogam- and annual-dominated vegetation on calcareous and ultramafic rock outcrops	Open pioneer grasslands with perennial succulents and spring annuals with subordinate small tussock grasses on calcareous outcrops
	R16	Perennial rocky grassland of Central and South-Eastern Europe	Open grassland generally dominated by perennial grasses with rich mixtures of associated rosette herbs, mat-formers and geophytes on both siliceous and calcareous bedrocks
	R18	Perennial rocky calcareous grassland of subatlantic-submediterranean Europe	Open grassland dominated by perennials and especially rich in mat-formers on calcareous bedrock
	R1A	Semi-dry perennial calcareous grassland (meadow steppe)	Semi-natural grassland dominated by mixtures of graminoids and forbs, often extremely species-rich and sometimes striking contingents of orchids on calcareous bedrock
	R1B	Continental dry grassland (true steppe)	Steppe and steppe-like grassland dominated by plants adapted to long periods of summer drought, mostly tall tussock grasses and perennial forbs on mostly limestone substrate
	R1M	Lowland to montane, dry to mesic grassland usually dominated by <i>Nardus stricta</i>	Semi-natural grasslands dominated by the tightly tussocky <i>Nardus stricta</i> on siliceous bedrock
ALPINE AND SUBALPINE GRASSLANDS	R41	Snow-bed vegetation	Dominated by grasses, sedges, herbs and cryptogams developed beneath late-lying snow patches in Arctic and subarctic lowlands, boreal mountains and temperate high mountains of Central and Southern Europe
	R43	Temperate acidophilous alpine grassland	Grassland and dwarf chamaephyte vegetation, typical of the highest summits and ridges, on siliceous bedrock
	R44	Arctic-alpine calcareous grassland	Grassland and dwarf chamaephyte vegetation, typical of the highest summits and ridges, on siliceous bedrock

Data collection and processing

We compiled a dataset of occurrence data in Europe of the 167 diagnostic species amounting at 2,613,066 points of occurrence. We relied on species occurrence data collected from four online databases: sPlot (Sabatini et al., 2021), Silene (silene.eu), GBIF (GBIF.org) and EVA (Chytrý et al., 2016). The data point was cleaned from any interferences such as data point falling within a Country's capital radius as well as those assigned to institutions (e.g., botanical gardens, herbaria, museum, etc.), poorly geo-referenced occurrences, duplicates records, coordinates outside the landmass and/or inside urban areas with the *clean_coordinate* function in the CoordinateClear R package (Zizka et al., 2023). We then retained for each taxon only one occurrence per 1x1 km grid cell to mitigate pseudo-replication of occurrences. The final dataset consisted of 2,497,067 occurrences.

Environmental predictors

We downloaded 19 bioclimatic variables for current (1981-2010) and future (2071-2100) time periods at about 1x1 km spatial resolution from the CHELSAv1.2 dataset (Karger et al., 2017). For the future scenario, we chose two representative concentration pathway (RCPs) representing possible future greenhouse gas emission trajectories and coded according to a possible range of radiative forcing values in the year 2100 relative to preindustrial values (+ 2.6 and + 8.5 W/m², hereafter RCP370 and RCP585, respectively). For each RCP we used projections from five internationally recognized general circulation models (GCMs): CESm1-CAM5, FIO-ESM, IPSLCM5A-MR, MIROC5 and MPI-ESM-MR. GCMs represent physical processes in the

atmosphere, ocean, cryosphere and land surface. We performed a principal component analysis (PCA), with the *dudi.pca* function implemented in the ade4 R package (Dray et al., 2025), on the 19 bioclimatic variables for current and for each future period (i.e., all the combinations of RCPs and GCMs) pooled together and kept the first two axes to reduce multicollinearity between predictors in order to minimize model overfitting. We integrated six soil variables by taking data on bulk density of the fine earth fraction, volume fraction of coarse fragment (>2mm), clay, sand, soil organic carbon in fine earth (SOC) and pH (H₂O) at about 1x1 km resolution from the SoilGrid database (<https://www.isric.org/>). As well for edaphic predictors, we performed a PCA analysis on the six soil variables and retained the first two axes to reduce model overfitting. Finally, to account for the topographic variability and heterogeneity in the model, we calculated the Topographic Roughness Index (TRI) from digital elevation models (DEMs).

Species Distribution Models

The species occurrence data and the five environmental variables were used for the construction of species distribution models (SDMs) for each selected diagnostic species. To account for model-based uncertainties in the modelling process, we used five species distribution modelling techniques implemented in the R package biomod2 v 4.2-6-2. (Thuiller et al., 2025), and these are: generalized linear model (GLM), generalised boosted models (GBM), artificial neural network (ANN), functional discriminant analysis (FDA), and maximum entropy (MaxNet). We

generated ten replicate sets of pseudo-absences for each species. For each pseudo-absence set, we repeated ten times a split-sample cross-validation, using a random subset (30%) of the initial dataset. We used two different measures implemented in BIOMOD2 to evaluate model performance: the Relative Operating Characteristic (ROC) and the true skill statistic (TSS). Models were calibrated on species distribution in Europe. We then averaged projections from different model techniques and GCMs to implement an ensemble forecasting approach and crop them on the Alps.

Habitat Suitability Framework

We used the resulted values from SDMs of habitat suitability to calculate the mean difference per grid cell of habitat suitability from the present to the future climate scenarios (for both RCP370 and RCP585) of each diagnostic species. This value indicates the magnitude of the in average habitat suitability change per grid cell. Negative values indicate an overall loss in habitat suitability with respect to the gain from present to future, whereas positive values indicate an overall gain in habitat suitability. Then, we calculated the percentage of cells that increase (positive percentage) or decrease (negative percentage) their suitability from present to future climates. Therefore, we implemented a new framework, called Habitat Suitability Framework, representing the indices together in a two-dimensional structure (Fig. 2). If species fall in the bottom left quadrat, it indicates that there is a general loss of habitat suitability respect to the present climate due to more cells losing suitability and an average loss of suitability per cell. On the contrary, it indicates that there is a

general gain of habitat suitability respect to the present climate due to more cells gaining suitability and an average increase in suitability per cell. If species fall within the top left quadrants, it means that more cells lose suitability but the average gain per cell is higher than the loss, that is, few cells gain suitability, but they gain a lot. If species fall within the bottom right quadrants, it means that more cells gain suitability but the average loss per cell is higher than the gain, that is, few cells lose suitability

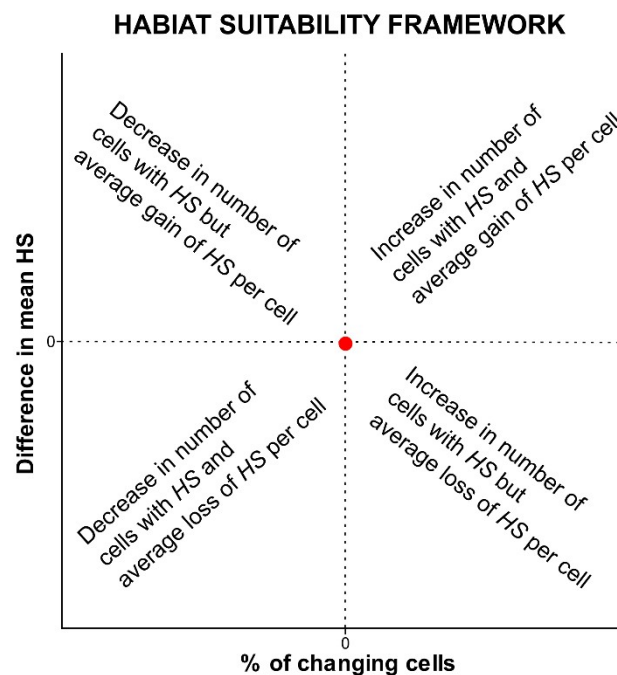


Figure 2

but lose a lot. Ultimately, if species are positioned near the zero intersection, it indicates that any change in habitat suitability is negligible.

Species Spatial shift and prioritization map

To account for the geographical shift in habitat suitability of each diagnostic species, and this is the eastern-western and/or the southern-northern spatial change, we

weighted the coordinates of the centroids of grid-cells per the value of future habitat suitability of that same cell.

We then generated prioritization maps by representing cold and hot zones indicating a pronounce loss or gain in habitat suitability, respectively, under future climates for the diagnostic species of each selected grassland habitats. For this, we calculated total habitat suitability per habitat as the sum of the habitat suitability values from each diagnostic species (Calabrese et al., 2014). We classified grid-cells according to the upper and lower decile of current and futures (RCP370, RCP585) stacked habitat suitability values per habitat. Then, we summed current and future decile for each grassland so that values equal or higher with respect to current values indicate a stable or gain in habitat suitability (i.e., hotspots) whereas values lower than current values indicate a loss in habitat suitability in the future (i.e., cold spots).

Species' tolerance range

Species capacity to face and endure changes in climates is known to depend on their physiological tolerance (Williams et al., 2006; <https://doi.org/10.1098/rsbl.2006.0541>). Therefore, we determine whether a correlation between species' tolerance ranges and changes in their habitat suitability exist. We first calculate species' tolerance with the *omi* function in the *ecospat* R package. Then, we used linear regression models to determine the relationship between species niche and future changes in climate with *lm* function in the *lmer4* R package.